Mathematics of Electoral Systems

Models and mathematical results
in the Euro-Atlantic world

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From ballots to political representation

One electorate. Different mathematical maps. Different winners.

Single-winner rules

Plurality (US, UK, Canada)
Runoff (France)
Indirect election (US presidency)

Multi-winner rules

Party-list PR (Spain, Romania, EU)
STV (Ireland, Northern Ireland)
Compensatory mixed systems (Germany)

Social choice theory

Majority cycles
Strategic voting
Impossibility results

Key distinction: parliamentary seat allocation ≠ aggregation of complete preference rankings.

Plurality: First-Past-the-Post

Model

District d elects one candidate.

$w(d) = \arg\max_i v_i(d)$

A plurality suffices; >50% is not required.

Illustrative result

A: 40% B: 35% C: 25%

A wins 100% of the district seat with 40% of votes.

Across districts, seat share may diverge strongly from vote share.

Used for lower-house district contests in the US, UK and Canada.

Two-round majority: France

Model

Round 1: test whether a candidate wins an absolute majority.

If not, eligible candidates proceed to round 2.

Winner = plurality of votes in decisive round.

Illustration: top-two runoff

Round 1: A 40%, B 35%, C 25%

If 80% of C voters switch to B:
B = 35 + 20 = 55%
A = 40 + 5 = 45%

B wins, unlike plurality.

The numeric example is top-two. French legislative runoffs may admit more than two candidates, subject to eligibility rules.

Party-list proportional representation: divisor methods

D’Hondt (Jefferson)

For each party i, generate quotients:

$q_{i,k} = v_i / k, \quad k = 1, 2, 3, \ldots$

Assign S seats to the S largest quotients.

Sainte-Laguë (Webster)

For each party i, generate:

$q_{i,k} = v_i / (2k - 1)$

D’Hondt generally favors larger parties more than Sainte-Laguë.

Party-list PR appears in Spain, Romania and European Parliament elections; exact formulas, thresholds and districts vary.

Same votes, different parliamentary seats

Party	Vote share	D’Hondt	Sainte-Laguë
A	48%	4	3
B	28%	2	2
C	14%	1	1
D	10%	0	1

D’Hondt can exclude a smaller party that gains a seat under Sainte-Laguë.

Hypothetical votes, no legal threshold; quotients tie at 14, but both receive seats; allocation is unaffected .

Mixed-member proportional representation: Germany

Two ballots, two roles

First vote: preference for a constituency candidate.

Second vote: party-list choice, determining overall party seat entitlement.

Mathematical principle

Party entitlement roughly follows proportional allocation of second votes.

Constituency winners are accommodated within that entitlement, subject to the applicable rules.

German 2025 reform: a constituency plurality winner from a party needs second-vote coverage; a district win alone is not sufficient.

Germany: Second-vote coverage, six district wins, four seats

Illustrative party A in one land: 4 seats assigned by the second-vote allocation; 6 constituency plurality winners.

District	A candidate	First-vote share	Seat awarded
1	A1	48%	YES
2	A2	45%	YES
3	A3	41%	YES
4	A4	39%	YES
5	A5	36%	NO
6	A6	32%	NO

4/6 plurality winners are seated; the other 2 receive no mandate. No extra seats are added.

Ireland: Single Transferable Vote, ranked ballots and transfer of votes

Droop quota

For V valid votes and S seats:

Q = floor(V / (S + 1)) + 1

Example: V = 1,000; S = 3
Q = floor(1000 / 4) + 1 = 251

Counting algorithm

1. Voter casts list of preferences.
2. Count first preferences.
3. Elect candidates reaching quota.
4. Transfer surplus vote value fractionally (or according to local rules).
5. Eliminate weakest candidates and transfer ballots.
6. Continue until seats are filled.

STV is preferential and multi-winner; transfer details differ by jurisdiction. It is not equivalent to list PR.

Ireland: STV, how a surplus can elect the next candidates

Hypothetical 3-seat constituency · 1,000 valid ballots · Droop quota = floor(1000/4) + 1 = 251

Initial first preferences

A	**400**
B	**250**
C	**200**
D	**150**

A elected; surplus = 400 − 251 = 149

Fractional surplus transfer

1 A elected: t = (400 − 251)/400 = 0.3725

100 votes → B: 37.25; 300 votes → C: 111.75

2 B = 287.25; C = 311.75 → C elected first.

C surplus = 60.75; all transferred to B (assumption).

3 B = 287.25 + 60.75 = 348 → B elected.

Elected in order: A, C, B. Fractional inclusive transfer is illustrative; Irish counting rules vary.

US Electoral College: weighted aggregation

State-level model

Each state s awards a number E_s of electoral votes.

In most states, a plurality winner receives all E_s votes.

National electoral tally: $T_i = \Sigma_s E_s \cdot I(w(s) = i)$

Mathematical consequence

The national popular-vote winner need not win the Electoral College.

Aggregation takes place after state-level winner-take-all mappings.

Hence national plurality and electoral-vote maximisation are distinct objectives.

Maine and Nebraska use district-based allocation for some electoral votes; 270 of 538 is the current majority threshold.

Condorcet cycles and Arrow's theorem

A majority cycle (13 / 18 century)

Pairwise majority: Candidate A defeats candidate B if more voters prefer A to B than B to A.

Condorcet winner: a candidate who defeats every other candidate in pairwise comparisons.

3 voters; 3 alternatives:

1: $A \succ B \succ C$
2: $B \succ C \succ A$
3: $C \succ A \succ B$

A beats B, B beats C, C beats A (all 2-1 scores).

Arrow (1951)

When a society has **at least three options**, no voting system can always turn people's individual rankings into one consistent social ranking while satisfying a set of reasonable rules.
Unrestricted choice: People can have any ranking of the options.
Unanimity (Pareto): If everyone prefers A to B, society must rank A above B.
Independence of irrelevant alternatives: Society's ranking of A vs B should depend only on how people rank A vs B, not on their views about a third option.
No dictator: No single person's rankings should always determine society's ranking.
Consistency: Society's overall ranking should be coherent and transitive.

Cycle is a counterexample to transitivity of pairwise majority, not a proof of Arrow's theorem.

Strategy, fairness, and measurement

Gibbard–Satterthwaite (1973)

For ≥3 alternatives and strict rankings:
deterministic: has only one winner
onto: every candidate can win at least for one profile
non-dictatorial: no single voter determines the winner
unrestricted domain: any preference profile is admissible
Every such voting rule is strategically manipulable: for some preference profile, a voter can obtain a more preferred outcome by misreporting preferences.

Gallagher disproportionality index (1991)

For party vote shares v_i and seat shares s_i (percent):

$$LSq = \sqrt{[\,½\,\Sigma_i\,(s_i - v_i)^2\,]}$$

LSq = 0 means exact proportionality; higher values imply greater disproportionality.

No election rule satisfies every desirable criterion; state the domain and axioms before judging a result.

Suggested problems

1 · Simulation

Generate voter preferences and districts. Compare FPTP, runoff and list PR.

When can a party win fewer votes but more seats?

2 · Algorithms

Implement D’Hondt, Sainte-Laguë and STV.

Study complexity, tie-breaking and sensitivity to party thresholds.

3 · Axioms

Construct a Condorcet cycle.

Investigate manipulation and monotonicity using a small exhaustive search.